b. Salt Stress

Presence of high salt concentration in the soil restricts the growth and development of plants. Most commonly the plants which are present near the seashore and estuaries are subjected to salt stress. According to an estimate about one third of irrigated land on earth is affected by salt stress. Na^{1}, Cl^{2}, K^{1}, Ca^{11} and Mg^{11} ions usually contribute to soil salinity. Plants growing in such areas face two problems:

1. Absorption of water from the soil with negative water potential
2. Interaction with high concentration of toxic sodium carbonate and chloride ions.

On the basis of salt tolerance, they are grouped into two categories:

1. Halophytes
2. Non-halophytes or glycophytes

Halophytes are native to saline soils. The halophytes which can resist a range of salt concentration are called as **euryhaline** and those with narrow range of resistance are called **stenohaline**. Non-halophytes cannot resist salts as the halophytes. *Helianthus annus* tolerates high Mn^{21} ions. Those which are present in salt regions face two problems:

- One is high concentration of salts in soil water leads to decrease in water potential so they grow in opposite direction. Example: *Salicornia.*
- Injuries in salt affected plants caused by both osmotic effects and specification effects. Accumulation of chloride ions reduces water absorption and transpiration.

Salt stress due to deficiency of mineral elements (K, P, S, Fe, Mo, Zn, Mg, Mn) causes physiological disorders which lead to reduced growth and yields.

1. Salt accumulators absorb and store salts so that the osmotic potential of their cells continues to remain negative throughout the growing region.
2. In some salt hardy plants, the excess salt is excluded on the surface of leaves. Some plants have salt glands which secrete salt (mostly NaCl). The exuded salt absorbs water hygroscopically from the atmosphere.
3. Some plants lose their excess salt by leaching into the soil or by dropping their salt filled leaves.
4. Salt tolerant plants (true halophytes) synthesize large amounts of the amino acid proline, galactosyl glycerol and some organic acids which function in osmotic adjustments.

Mechanism of salt tolerance

The plants growing in salty habitats like halophytes face the problem of excessive dissolved salts in the solution. Excess of salt creates comparatively more negative osmotic potential so that the plants tend to lose water into surrounding medium. Under such conditions the plants tend to lose water only when their water potential becomes more negative. It is possible only if they absorb excess of salt and accumulate it in their cell saps to maintain the same or higher concentrations as those of outside plants.

The drawbacks:

1. Salt accumulates in the vacuoles
2. The plants become succulents
3. Accumulated salt dehydrates the cytoplasm
4. Sodium chloride cannot be tolerated in the cytoplasm and it denatures several enzymes

Thus, absorption and accumulation of inorganic salts fail to solve the problem. The plants however tolerate the salt stress by synthesizing organic compounds that can exist at high salt concentrations without denaturing the enzymes. These organic compounds are called **nontoxic organic osmotica**. Examples: Proline and Betalin (osmoregulators).

Summary

Growth occurs by cell division, cell elongation and cell maturation. The first phase is lag phase, the second is log phase and the final phase is steady state phase. The

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